

M003 Week 1 — English Worksheet

Topic: First Steps with Coordinates (x, y) · **Course:** 3D Coordinates · **Time:** about 45 minutes

This week you draw on a flat **wall**. Every spot has a name — **two numbers: (x, y)**. x is how far **across**, y is how far **up**. Counting starts at **0**.

1 · Predict 🧠

Read each set of steps. Write what you think you will see.

```
place gold block at (3, 0)
place gold block at (3, 5)
```

The two gold blocks share the same x. Are they close together or far apart? Which number changed?


```
place red block at (0, 0)
place red block at (4, 0)
place red block at (0, 4)
```

Three red blocks. Which one sits at the corner where counting starts? Which one is higher up the wall?


```
set y to 2
place gold block at (0, y)
set y to 6
place gold block at (0, y)
```

The variable y changes from 2 to 6. Where does each gold block land? Do they share the same x ?

2 · Spot the Bug 🐛

Each code block below is broken. Read what it should do, fix it, then explain why the original was wrong and your fix works.

Bug A — The block should land at **(2, 5)** — 2 across, 5 up. Right now it lands elsewhere.

```
place gold block at (5, 2)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The block should sit at the **bottom corner** of the wall. The coder forgot where counting starts.

```
place gold block at (1, 1)
```

Hint: counting starts at **0**, not 1.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The block should sit **5 up** at $x = 0$ — that is **(0, 5)**. Right now it drops below the wall.

```
place gold block at (0, -5)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Now switch to your homework world. Place a gold block at **(0, 0)** so you can see where counting starts. Then pick three more coordinates, place a different block at each one, and write down the coordinates you used. When you finish, come back here.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.